Types of Security

No security

Basic Security (id and password)

Security using obsecurity (Hiding the data)

Security through Encryption

Principle of Security / Security Services

C I A

- confidentiality
- Integrity
- Availabality
- Non Repudiation
- Authentication (verification)
- Access Control

Role management

Rule management

Security Attaks

Possive Attak

Active attack

Host A

Sender

Host B

Reciever

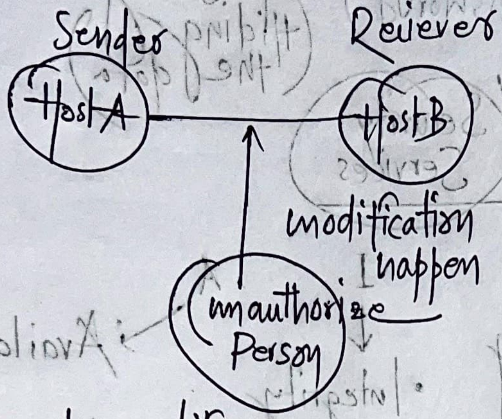
unauthorise person

man in the middle attack

Passive Attacks

- Attackers does not do any modification, only try to see message
- Finding Attacker is more difficult than active attack.
- Trial and Error method is solution

Active attack



- Attacker try to modify message
- Finding Attacker is less difficult than passive attack
- detection is solution.

cryptography

The art/science of achieving security through encryption

encryption → Encryption is method / process through which plain text is convert into cipher text

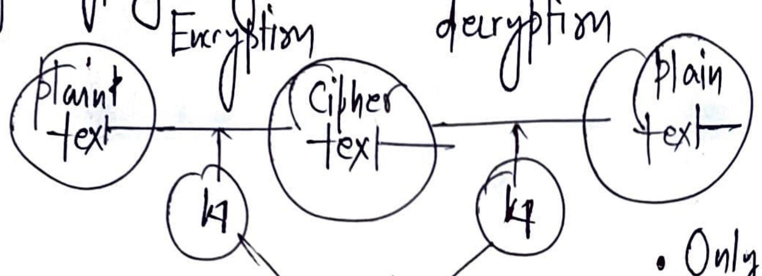
Key based Cryptographic Algorithm

2. Assymmetric key Cryptography

Example of Algorithm

1. Symmetric key Cryptography → Data Encryption Standard (DES) Algorithm

- Also known as private key cryptography



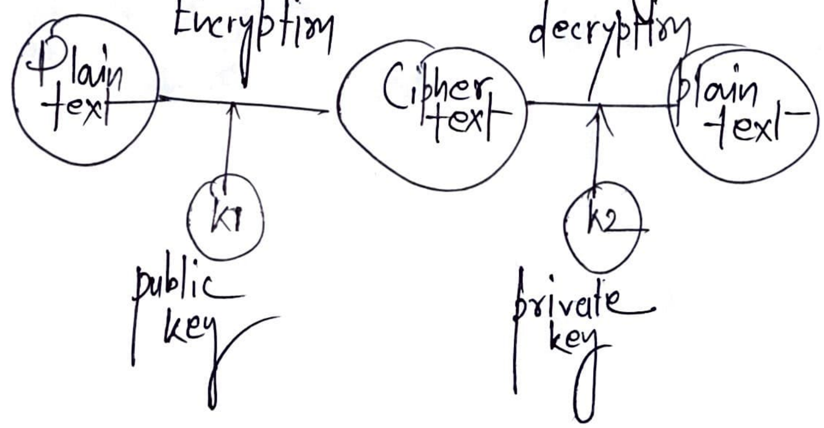
- Only private

Same keys are used for both Encryption & decryption

Assymetric key Cryptography

- public key Cryptography
- used in RSA and DSA algorithm

Here we use two different keys (i) public and (ii) private keys



Hash code

Hash code uses no keys. It uses some function which is called as Hash function. Varriable length

Ceaser Cipher Algorithm: → Symmetric key algorithm

- First algorithm in sequrity field
- also known as Monoalphabetic Cipher

Example

C E A S E R

if key = 3 +3 +3 +3 +3 +3 +3

F H D V H U

Monoalphabetic

plaintext's one letter is converted into only fixed one letter every time

polyAlphabetic

Example-2

Meet me at two PM.

key = 4

M E E T ME AT TWO PM

+4

Q I I X QI EX XAS TQ

cipher text → (QIIXQIEXXASTQ)

For large question

Encryption:

cipher text (c)

$= E(p,k) = (p+k) \bmod 26$

Decryption:

plain text $(p) = D(c,k) = (c-k) \bmod 26$

	A	B	C	...	Z
Alphabetic order	0	1	2		25

$1 \le k \le 25$

Plain text → ~~HELLO~~ Hello key = 4

cipher(H) = E(H,4) = (7+4) mod 26
= 11 % 26 = 11 = L

Cipher(E) = E(E,4) = (5+4) mod 26
= 9 mod 26 = I

cipher(L) = E(L,4) = (11+4) mod 26
= 15 % 26 = 15 = P

cipher(L) = P

cipher(O) = E(O,4) = (14+4) mod 26
= 18 % 26 = 18 = S

cipher text of (HELLO)
→ C(LIPPS)

Technique of Cryptography

Technique of Cryptography

- Substitution
  - HELLO → LIPPS
  - Substitute the alphabet by another alphabet
- Transposition
  - Change the position of the alphabet
  - CEASER — Plaintext
  - ASEERC — Cypher
  - key Based
  - keyless

Transposition Technique

1. Rail Fence Technique → keyless transposition

Plaintext → MEET ME AT COLLEGE

No of Row → 2 (Fixed)

1st Row → M E M A C L E E

2nd Row → E T E T O L G

Number of columns depends upon the length of the input text.

cipher text

(MEMACLEEETETOLG) → In cipher text there should not be any space

Row Transposition / simple Columnar Technique → (keyed Transposition)

Plaintext → meet me at college

key ← integer type (Example — 41235, 51234 / 164325 etc)

key given → CRYPTO

C O P R T Y
↓ ↓ ↓ ↓ ↓ ↓
1 2 3 4 5 6

(Position of the alphabet)

C R Y P T O
1 4 6 3 5 2

1	4	6	3	5	2
M	E	E	T	M	E
A	T	C	O	L	L
E	G	E	X	Y	Z

indicates the column number

Dummy Bits / Bogus Characters

1st column → MAE
2nd Column → ELZ
3rd column → TOX
4th column → ETG
5th column → MLY
6th Column → ECE

cipher text
→ MAEELZTOXETGMLYECE

Double Columnar Technique

Plaintext → MEET ME AT COLLEGE
key → CRYPTO
(146352)

Step 1 do the process to convert pt to ct by simple Columnar Technique

cipher text
→ (MAEELZTOXETGMLYECE)
↓
This is the plain text of 2nd step

2nd step

1	4	6	3	5	2
M	A	E	E	L	Z
T	O	X	E	T	G
M	L	Y	E	C	E

1st Colum → MTM
2nd Column → ZGE
3rd Column → EEE
4th Column → AOL
5th Column → LTC
6th Column → EXY
ciphertext
→ (MTMZGEEEEAOLLTCEXY)

4 keyed Transposition

Plain Text
→ ATTACK POSTPONED UNTILL TONIGHT
key → (43152)

Step1 Break input text into key length size segment

ATTACK ← 1st segment
2nd segment → KPOST
3rd segment → PONED
4th segment → UNTIL
5th → TONIG
6th Segment → HTXYZ

key 4 3 1 5 2
↓ ↓ ↓ ↓ ↓
1 2 3 4 5

For ~~ever~~ every segment
4th alphabet → 1st alphabet
3rd alphabet → 2nd alphabet
1st alphabet → 3rd
2nd → ~~4~~5th | 5th → 4th

1st segment (ATTAC)
↓
ATACT

2nd segment (KPOST) → SOKTP
3rd seg (PONED) → ENPDO
(UNTIL) → ITULN
(TONIG) → INTGO
(HTXYZ) → YXHZT

cipher text
(SOKTPENPDOITULNINTGOYXHZT)

Substitution Method

Plain Text HOW ARE YOU?
key → NCBTZQARX

Plain Text	H	O	W	A	R	E	Y	O	U
	7	14	22	0	17	4	24	14	20
key	N	C	B	T	Z	Q	A	R	X
	13	2	1	19	25	16	0	17	23
Addition value	20	16	23	19	42	20	24	31	43
Subtract 26 from those which ≥ 26	20	16	23	19	16	20	24	5	17
	U	Q	X	T	Q	U	Y	F	R

cipher text — UQXTQUYFR

Decryption (Cipher text to Plain text)

Plain Text	U	Q	X	T	Q	U	Y	F	R
	20	16	23	19	16	20	24	5	17
Key →	N	C	B	T	Z	Q	A	R	X
	13	2	1	19	25	16	0	17	23
Subtract	7	14	22	0	-9	4	24	-12	-6
Add 26 which <0	+3	2							
	7	14	22	0	~~15~~ 17	4	24	14	20
	H	O	W	A	R	E	Y	O	U

Plaintext → (HOWAREYOU)

Playfair Cipher

Plaintext → SISTER NIVEDITA UNIVERSITY
key → CRYPTOGRAPHY

step1 break input text into pair of two
multiple segment
(Each segment consist of two alphabet)

~~key~~ SI ST ER NI VE DI TA
UN IV ER SI TY

step2 • if number of alphabet is odd then just add z

• same word in a pair

AA → AX XA

Step2 • Create a 5×5 matrix
• insert all the alphabet of key one by one.
* only one character/alphabet will inserted only one time. If the alphabet repeated then no need to re-insert it.
• Then insert all the ~~plain text~~ alphabet one by one
• I/J considered as a single alphabet

C	R	Y	P	T
O	G	A	H	B
D	E	F	I/J	K
L	M	N	Q	S
U	V	W	X	Z

Some Rules

1) If both letters are in the same row. take the letter to the right of each one (going back to the leftmost if the right most position end)

AB → HO

2) If both letters are in same column take the letter below each one (going back to the top if at the bottom)

FW → NY

3) If ~~neither~~ none of the above rules is true: form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

NZ → SW

SI ST ER NI VE DI TA UN IV ER SI TY
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
QK ZB MG QF RM EK YB WL EX MG QK CP

cipher text

(QKZBMGQFRMEKYBWLEXMGQKCP)

Data Encryption Standard (DES) / Data Encryption Algorithm (DEA)

→ Symmetric key Algorithm

64 bit plain text

key (56 bit) → DES Algorithm

64 bit cipher text

→ Every 8th bit will be discarded
Total = 8×8 = 64
64 − 8 = 56 bit

64 bit plain text
↓
64 bit → initial Permutation
↓
32 bit LPT | 32 bit RPT
↓
* 16 Rounds — 16×5 = 80
↓
Final Permutation (64 bit)
↓
64 bit Cipher text

* Each round have 5 steps that are

(i) key transformation
(56 bit key is transformed into 48 bit)

left circular shift

(ii) Expansion permutation → (32 bit → 48 bit)

(iii) S-box (Substitution box)

(iv) P-box (Permutation-box)
→ 32 bit

(v) XOR and SWAP

Step 1) key transformation

- 56 bit key is transformed into 48 bit using left circular shift

56 bits

28 bits

28 bits

4 bits discarded

4 bit discarded

24 bits

24 bits

48 bits

user choice

Round	Shift
1	1
2	2
3	2
⋮	⋮
16	2

minimum → 1 LS

max → 2 LS

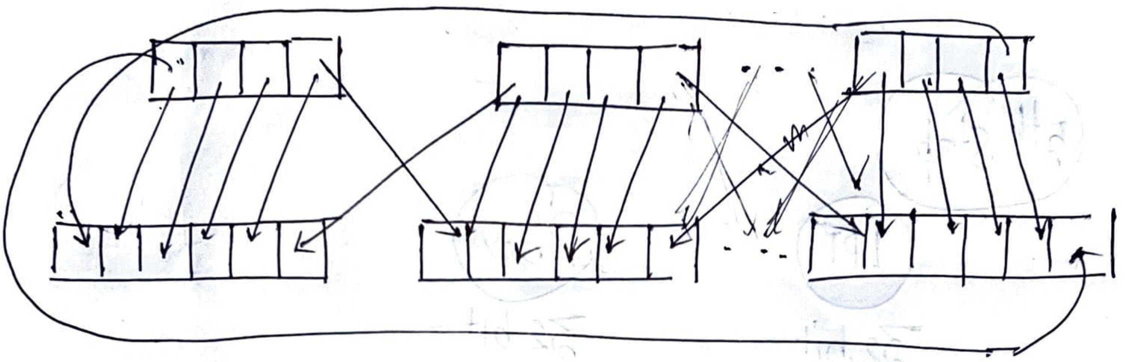
2nd step

Expansion permutation

It happens on RPT, Here 32 bit transformed into 48 bit

32 bit devided into 8 parts, which contains 4 bit each part

32 bit RPT



Substitution box

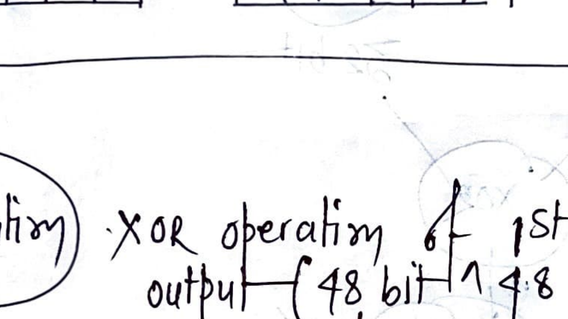
XOR operation of 1st & 2nd step output (48 bit ^ 48 bit)

↓

48 bits

devided into 8 parts
each parts → 6 bits

4 rows 16 columns
64 block



each box contains 0 - 15 (4 bits binary value)

We uses 8 boxes
Each boxes → 6 bit i/p

1 010010
10 → Row
0101 → column

2nd row and 5th column value will be 4th bit binary number which replace 6 bit (101010)

Total 8 Sbox

Each S box Output → 4 bit

Total O/P → $8 \times 4 = 32$ bit

P-box Juggle up

O/P → 32 bit

5th Step

LPT 32 bit

P box 32 bit

XOR

32 bit O/P

New RPT

Old RPT

New LPT

This called Swap

Diffie Hellman key Exchange Algorithm:

key should be primary number and number should be large

1) n, g prime numbers
   $n \longrightarrow$ Allice } known by both
   $g \longrightarrow$ Bob

2) Alice x: (random numbers any)
   calculate $A = g^x \bmod n$

3) Alice send A to Bob

4) Bob, y
   $B = g^y \bmod n$

5) Bob sends B to Alice

6) Alice $k_1 = B^x \bmod n$

7) Bob $\longrightarrow k_2 = A^y \bmod n$

Example $n = 11, g = 7$ | $x = 3$, $y = 6$

Advanced Encryption Standard (AES)

Rijndaet

proposed 2006

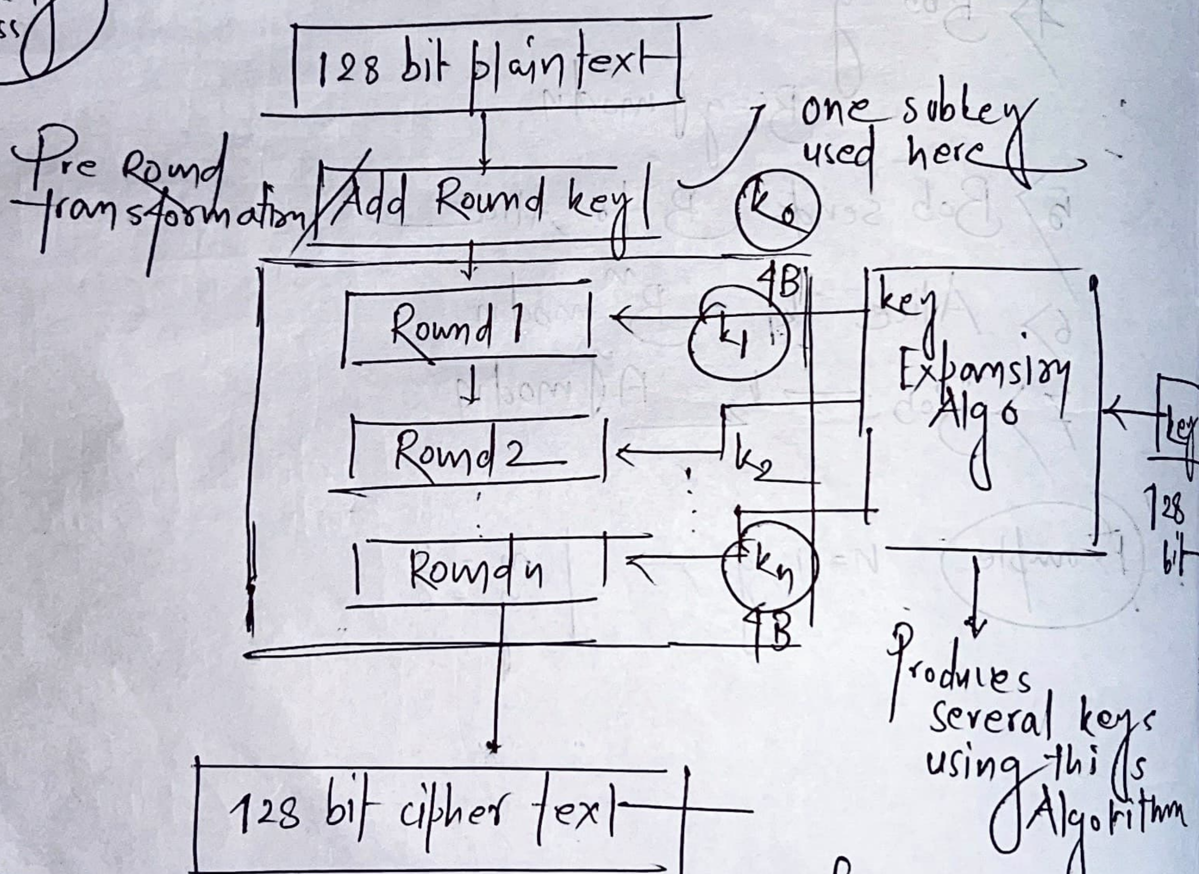
2001

128 bit plain text

128 bit cipher text

Rounds.	length of key	Version name
10	128 bits	AES 128
12	192 bits	AES 192
14	256 bits	AES - 256

Working process



Steps

1. Substitution Bytes
2. Shift Rows
3. mix columns
4. Add Round key

if 10 Round

(10+1) key

= 11×4B

= 44 B

proposed 2000

Advanced Encryption Standard (AES)

Rijndaet

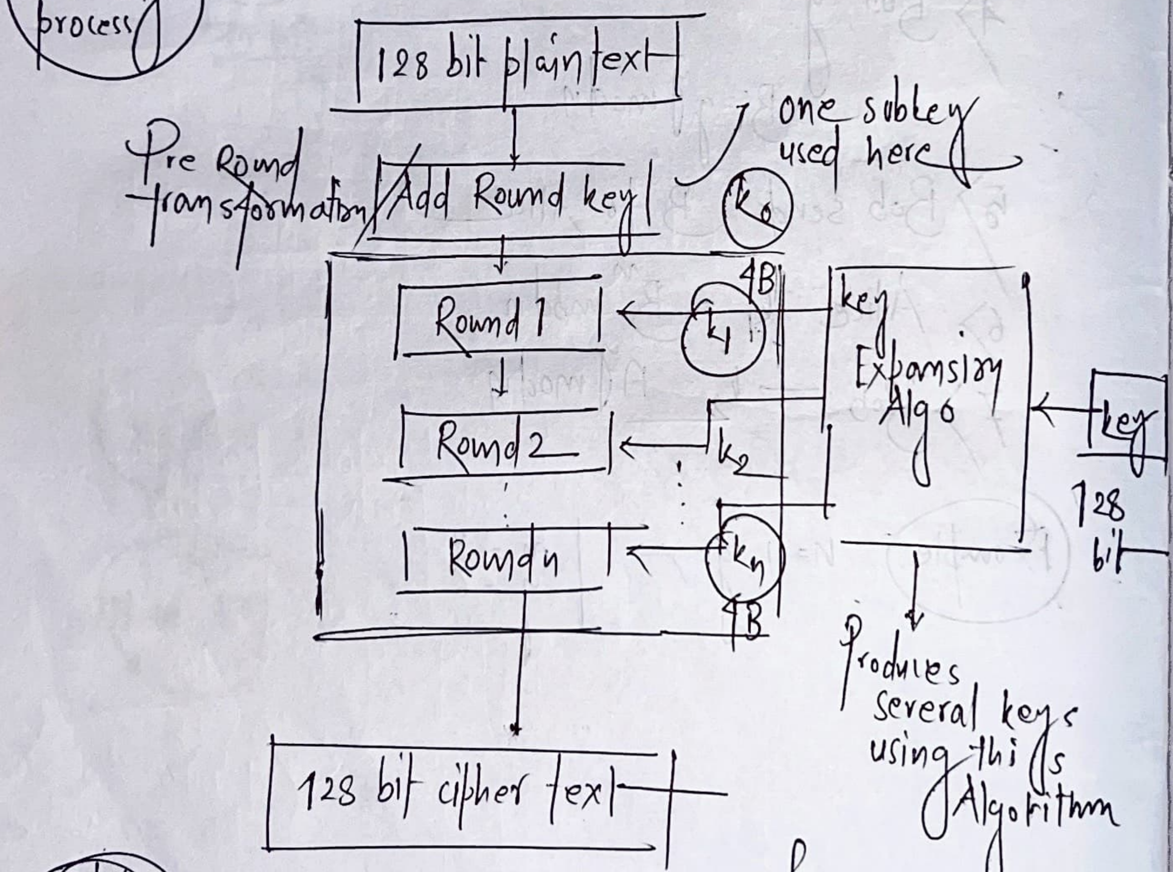
2001

128 bit plain text

128 bit cipher text

Rounds.	length of key	Version name
10	128 bits	AES 128
12	192 bits	AES 192
14	256 bits	AES 256

Working process



Steps

1. Substitution Bytes
2. Shift Rown
3. mix columns
4. Add Round key

if 10 Round

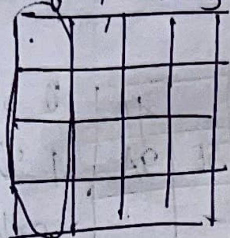
(10+1) key

= 11×4B

= 44 B

Word 128 bit

W0 W1 W2 W3



4x4

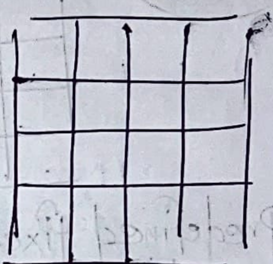
1 column = 1 Word
↓
4 Byte

Add round key

XOR operation of 1st subkey k0 and 128 bit plain text for 128 bit AES

128 bit

1B 1B 1B 1B

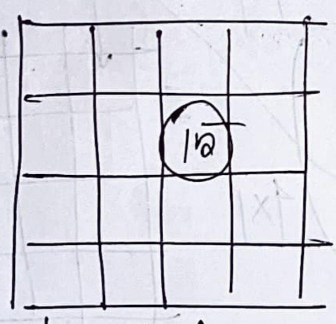


o/p of 1st step

Each word ~~32~~ bit
4 Byte → 32 bits
~~4 Byte~~

Values will be Hexadecimal

1.



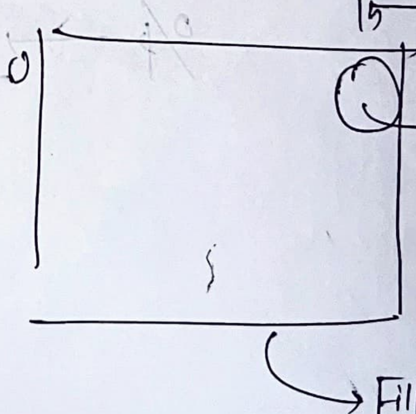
Hexadecimal format

0000 1111
Rows column

0th row and 15th column

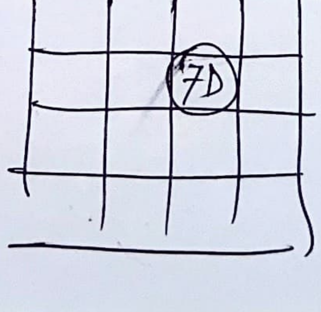
S-box 16X16

0 15



→ 7D

Fill with some value



2. Shift Rows

After 1st ~~Round~~ step the State matrix

0	3A	1D	2C	4D
1	9A	7D	4F	3E
2	5A	7C	FF	AD
3	4D	6D	4F	4D

i/p of 2nd step

$\xrightarrow{\text{left shift circular}}$

3A			4D	0 shift
7D	4F	3E	9A	1 shift
				2 shift
				3 shift

After 2nd step this is state matrix

3. mix columns

i/p state matrix

4x4

Here i/p matrix is multiplied with this predefined matrix
But 1 column at a time

Predefined fixed matrix

$$\begin{pmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{pmatrix}_{4\times4} \times \begin{pmatrix} \\ \\ \\ \end{pmatrix}_{4\times1}$$

1st column

o/p $\longrightarrow$ $\begin{pmatrix} \\ \\ \\ \end{pmatrix}_{4\times1}$

State matrix

4. Add Round key

state matrix of 3rd step [ ] 4x4 ⊕ [key (k_4)] 4x1 = [128 bit state matrix] 4x4

generated in key expansion Algo

4x1

Every Round key will change

This will be the i/p of next Round

Round 10th

The 3rd step (mix columns) will not be

Num of Rou → 10

Total subkey → 10+1 = 11

Each subkey → 4B

Total → 44B sub genrated by key expan Alg.

International Data Encryption Algorithm (IDEA)

Plain-text (64 bits)

Rounds → 8

key - 128 bits

input text (64 bits)

P1 | P2 | P3 | P4

Round 1 ← k1 … k6

Round 2

Round 3

⋮

Round 8

Output Transformation ← k49 … k52

64 bit Cipher text

input text
↓
4 parts each — 16 bit

Each Round 6 sub keys and consist of 14 steps

4 subkeys used

→ 4 parts
↓
cipher text (64 bit)

14 steps

1. Multiply
2. Add
3. Add
4. multiply
5. XOR
6. XOR
7. multiply
8. Add
9. multiply
10. Add
11. XOR
12. XOR
13. XOR
14. XOR

14 steps

1. multiply P1 and k1
2. Add P2 and k2
3. Add P3 and k3
4. multiply P4 and k4
5. XOR step1 and step3 output
6. XOR step2 and step4 o/p
7. multiply step 5 and k5
8. Add step6 and step7
9. multiply step8 and k6
10. Add step7 and step9
11. XOR step1 and step9
12. XOR step3 and step9
13. XOR step2 and step10
14. XOR step4 and step10

4 parts will be used as i/p of next Round

Rivest Cipher-5 (RC5)

Round → 0-255

keys → 0-255 → octate form 8/16/64

i/p text — devided into two parts

↓ word

↓ 2word i/b

16, 32, 64 bit

Steps

1. XOR
2. Circular left shift
3. Add with the key

Actual Steps

1. devide the plain text into two parts A and B (equal size)
2. Add A and subkey[0] producing → C
3. Add B and subkey[1] producing → D

Preround transformation

4. XOR C and D → E
5. circular left shift of E by D bits
6. Add E with Subkey[2i] → F
7. XOR E and F → G
8. Circular left shift G by F bits
9. Add G with Subkey[2*i+1] → H

Condition — No → C=F, D=H → (back to step 4)

Condition — yes → stop

Blowfish Bruce Scheneier → Symmetric key Algorithm

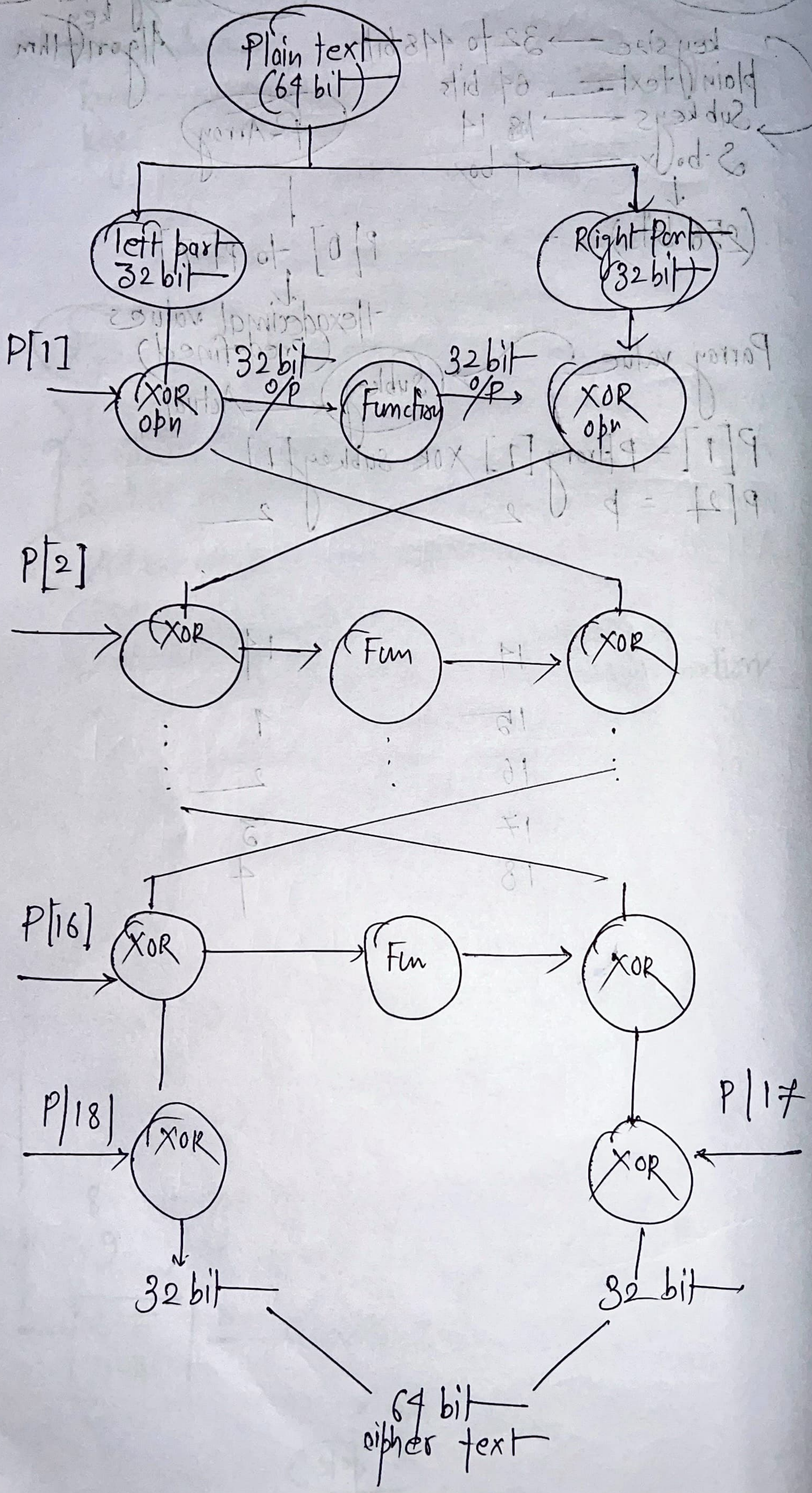
key size → 32 to 448 bits
plain text → 64 bits
Subkeys → ~~18~~ 14
S-box → 4 box
(256 bits)

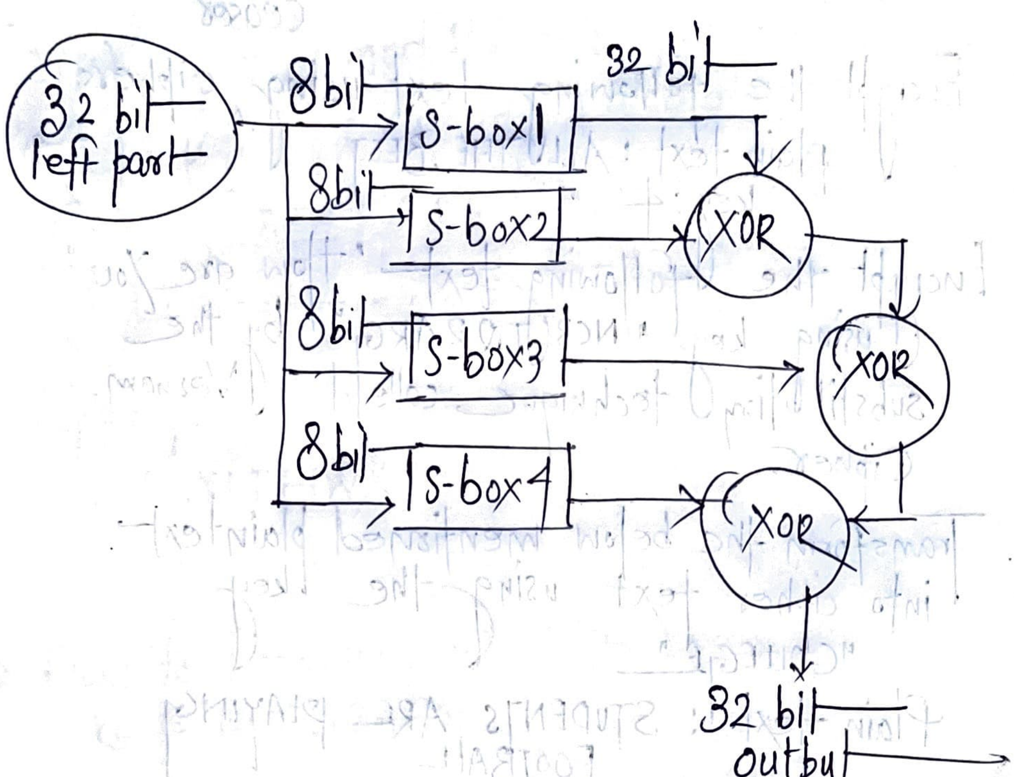
P-Array
P[0] to P[17]
Hexadecimal values
(Predefined)

Parray value → Subkey ← Actual

P[1] = PArray[1] XOR Subkey[1]
P[2] = P 2 2
:

14	14
15	1
16	2
17	3
18	4


Plain text (64 bit)
left part 32 bit
Right Part 32 bit
P[1]
XOR opn
32 bit o/p
Function
32 bit o/p
XOR opn
P[2]
XOR
Fun
XOR
P[16]
XOR
Fun
XOR
P[18]
XOR
P[17]
XOR
32 bit
32 bit
64 bit cipher text


32 bit
left part
8 bit
S-box1
32 bit
8 bit
S-box2
XOR
8 bit
S-box3
XOR
8 bit
S-box4
XOR
32 bit
output

ceaser

1. Encrypt the following text using ceaser ciphers:
plain text : ALL THE BEST
key : 4

2. Encrypt the following text "How are You" using key "NCBTQZARG" by the Vernam substituting technique called Vernam Cipher.

3. Transform the below mentioned plaintext into cipher text using the key "COLLEGE"
Plain text : STUDENTS ARE PLAYING FOOTBALL
by using Playfair cipher.

Solution 1

1) Plain text : ALL THE BEST
key = 4 $\therefore k=4$

For encryption,

$C(P) = E(P,4) = (P+4) \bmod 26$

$C(A) = E(A,4) = (A+4) \bmod 26$
$= (0+4) \% 26 = 4 = E$

$C(L) = (11+4) \bmod 26$
$= 15 \bmod 26 = 15 = P$

$C(L) = P$

$C(T) = (19+4) \bmod 26 = 23 \% 26 = 23 = X$

$C(H) = (7+4) \bmod 26 = 11 \% 26 = 11 = L$

$C(E) = (4+4) \bmod 26 = 8 \% 26 = I$

$C(B) = (1+4) \bmod 26 = 5 \% 26 = 5 = F$

$C(E) = \cancel{M}\ I$ (Already calculated)

$C(S) = (18+4) \% 26 = (22 \% 26) = 22 = W$

$C(T) = X$ (Already Calculated)

Cipher text

EPPX~~X~~LIFIWX

2) Plain text = HOW ARE YOU

KEY = NCBTQZARG

HOW	H	O	W	A	R	E	Y	O	U
	7	14	22	0	17	4	24	14	20
	N	C	B	T	Q	Z	A	R	G
	13	2	1	19	16	25	0	17	6
	20	16	23	19	33	29	24	31	26

Subtracting 26 from those which are greater than 26

20	16	23	19	7	3	24	5	0
U	Q	X	T	H	D	Y	F	A

Cipher text will be

UQXTHDYFA

3 **plain text**

STUDENTS ARE PLAYING FOOTBALL

key → "COLLEGE"

We have to break the plain text in pair of 2

ST UD EN TS AR EP LA YI NG FO OT BA LX XL

We are creating a matrix which contains the character in a cell at a time

C	O	L	E	G
A	B	D	F	H
I/J	K	M	N	P
Q	R	S	T	U
V	W	X	Y	Z

Rules

I) If two alphabet are in same Row, then ~~the~~ take the letter right side each one.

ST → TU

II) If two alphabet are in same column, then ~~ill~~ take the letter below of that each alphabet.

EN → FT

III) If any pair is not following Rule1 and Rule2 then create a rectangle and replace the alphabet or take the letter horizontal opposite corner.

Playintext —

ST UD EN TS AR EP LA YI NG FO OT BA L~~LX~~ XL

Cypher text —

TU SH FT UT BQ GN CD VN PE BE ER DB DL~~EA~~LD

cypher text

TUSHFTUTBQGNCDVNPEBEERDBDLLD

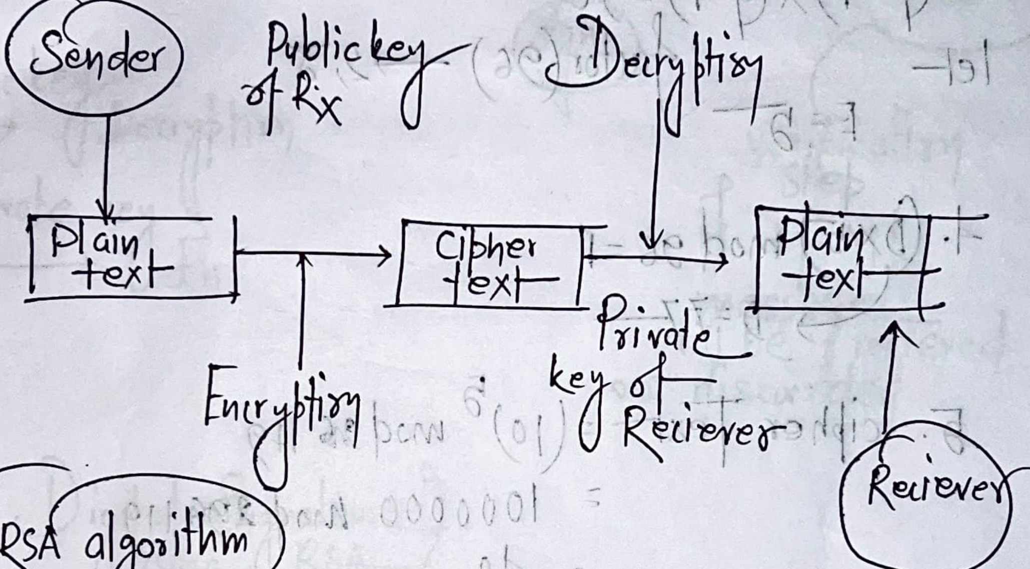
Public key Cryptography:

key used:
- Public key → Encryption key
- private key → Decryption key

Sender have public key of Reciever

Public key, Private key → Reciever

Plain text



RSA algorithm (Rivest - Shamir - Adlerman)

Steps

1) Take two large prime numbers, p & q (choose)

2) Calculate, $N = p \times q$

3) Calculate public key (Encryption key), E in such a way that the key is not any factors of $(p-1) * (q-1)$

4) Calculate private key (Decryption key), D, which satisfy the following ~~rule~~ condition

$(D * E) \bmod (P-1) * (q-1) = 1$

5> For encryption,

$$\text{cipher text}(CT) = (PT)^E \bmod \cancel{q}N$$

6> For decryption

$$PT(\text{Plain text}) = (CT)^D \bmod N$$

Example → Plain Text = 10

1. $p = 7 \quad q = 17$
2. $N = 7 \times 17 = 119$
3. $(p-1)\times(q-1) = 96$

let factor(96) $\longrightarrow$ 2, 3

$E = 5$

4. $(D \times 5) \bmod 96 = 1$

$\longrightarrow D = 77$

5. $\text{cipher Text} = (10)^5 \bmod \cancel{96}\ 119$

$= 1000000 \bmod \cancel{296}\ 119$

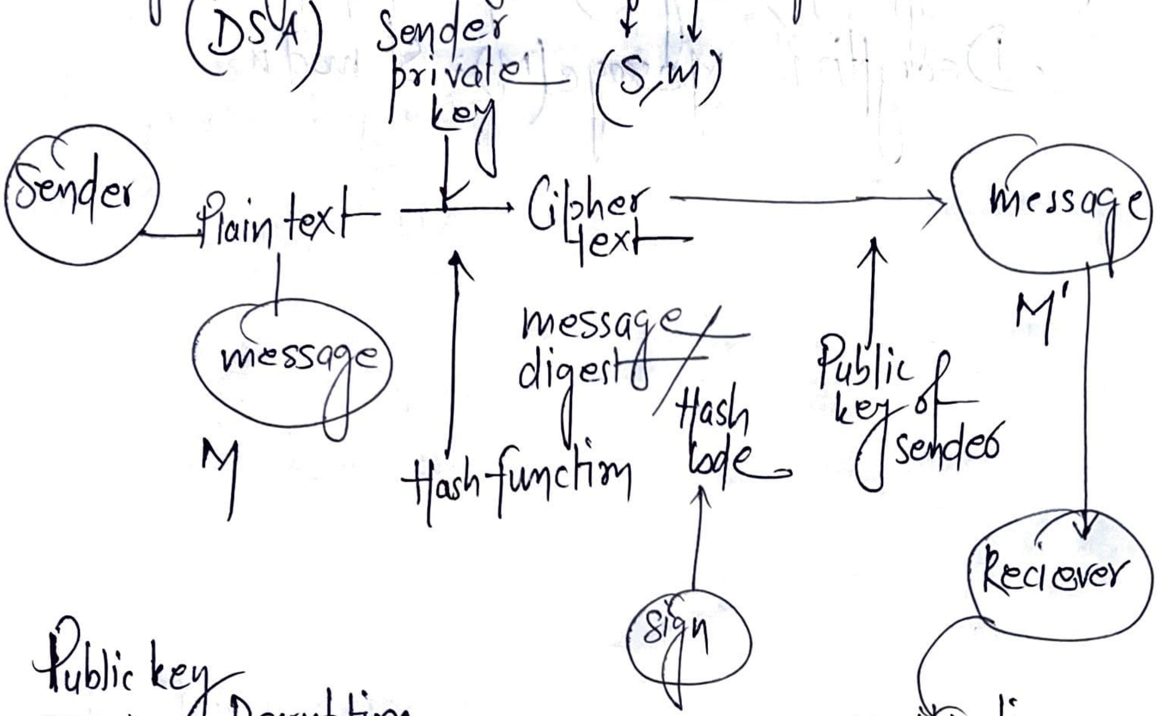
$= 40$

6> $PT = 40^{77} \bmod 119$

$= 10$

Digital Signature Algorith sign message

(DSA)



Public key $\longrightarrow$ Decryption

Private key $\longrightarrow$ En

verification step

If $M = M'$

message will be recieved or discarded

Digital Signateure using RSA

5) Encryption $\Rightarrow$ Sign(S) = $M^D \bmod n$

Decryption Message(M) = $S^E \bmod n$

Symmetric key	Assymmetric key
1. This is also known as private key or Secret key Cryptography	1. This is also known as public key Cryptography
2. Only one key is used for both encryption and decryption.	2. Two different ~~key~~ ~~is~~ (public & private) keys are used for encryption and decryption Respectively.
3. This is faster ~~for~~ in ~~ex~~ execution.	3. This is slower in execution.
4. It is less complex & less computational power is require	4. It is more complex and more computational power is required.
5. It is used for the transfer of bulk data (because it executes faster)	5. It is used for secretly exchanging the secret key.
6. Sharing the key b/w Sender & Reciever is not safe.	6. No problem of key sharing because of private key concept
7. Commonly used algorithms are DES, AES, RC5 2DES, 3DES etc	7. Common used algo are RSA, DSA etc

DES Algorithm Analysis:

- **Avlanche Effect**

A small change in plain text or key ~~can~~ should create a significant change in the cipher text

- **Completeness effect**

~~If~~ Each bit of the cipher text needs to depends many bits of the plain text.

Disadvantage of DES

> **key size**: In DES 56 bit key is required for encryption. Hence a total of 2^{56} combinations can be made out of these 56 bit keys. In todays parallel processing it is very easy two crack the actual key

> **Weak keys**: There are 4 weak keys
- All 0's
- All 1's
- Half 0's
- Half 1's

Semi Weak keys: Six pair of keys are called semi weak keys

possible weak keys: There are 48 poss weak keys out of 2^{56} ~~comb~~ Combination

Key Classtering

It means that 2 or more keys can create the same cipher text from the plain text.

Weakness in cipher design

Two specifically choosen i/p two S-box arrays can create same o/p

Hills cipher

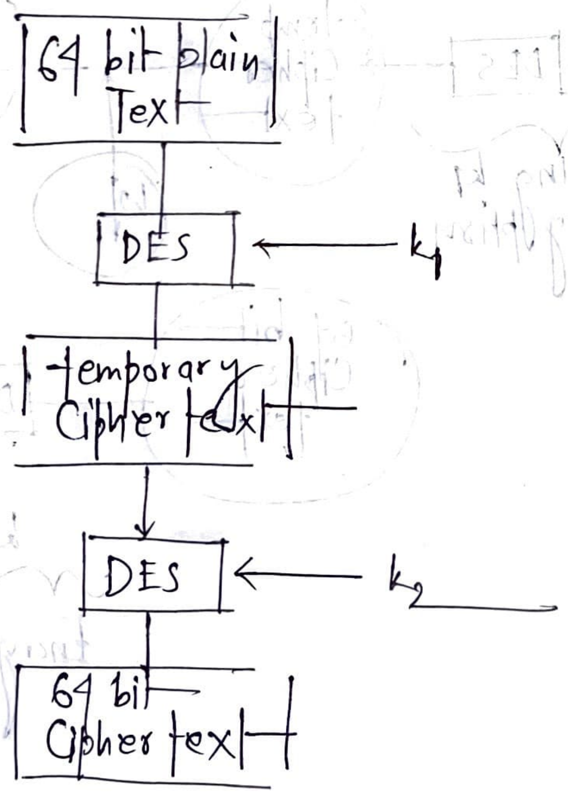
~~Types of Attacks:~~

- Virus
- Worm
- Frozen Horse

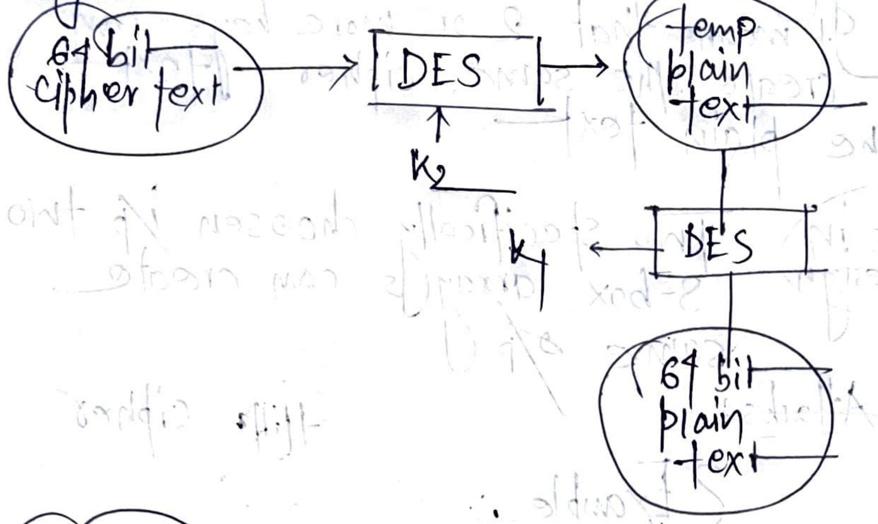
Example of Attack

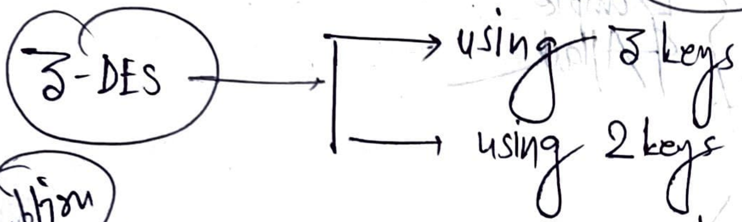
Double DES

Encryption

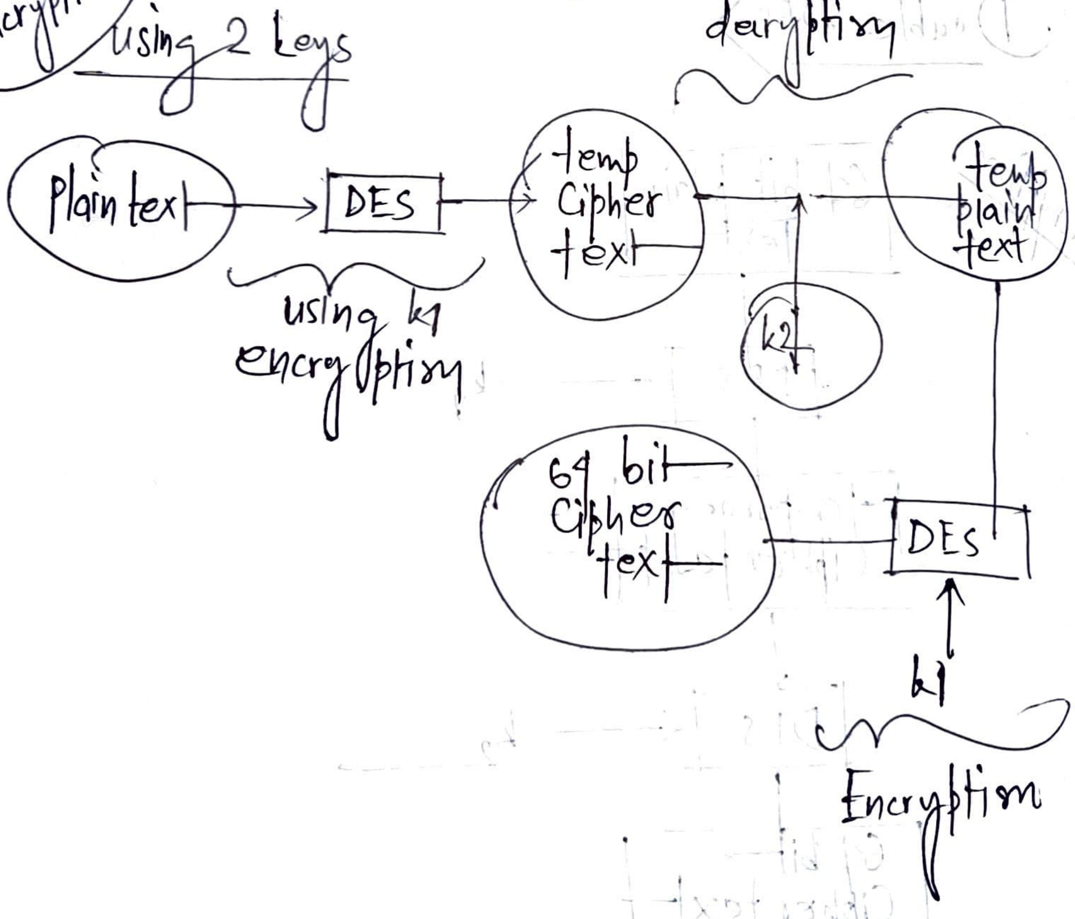


Decryption





Encryption using 2 keys



using 3 keys